XR Event Planning

Testing Plan for Interactive Prototype 2b
DECO2300

1. 3-Sentence Pitch

XR Event Planning is a VR application that lets users arrange and customize an event venue from inside the space. Users can add, move, rotate, and remove furniture, adjust lighting, and change themes through spatial interactions. This iteration introduces a Welcome screen, starter furniture, Sofa and Plant options, and environmental details to investigate how users begin planning and adapting an existing venue.

2. Testing Objective

Previous IP2 testing recorded 5/5 independent completions for furniture arrangement, removal, lighting, and theme changes. However, P2 described the venue as empty, and P3 linked its emptiness to uncertainty about where to begin.

This test examines whether participants can start independently and make spatial decisions while adapting a furnished venue. It also investigates physical manipulation, control discovery, and guidance using the Meta Quest headset. Previous IP2 testing used the Unity XR Interaction Simulator.

3. Assumptions and Unknowns

The test will examine whether:

- **A1: First-entry orientation.** Participants can discover Start and begin a relevant planning action without facilitator guidance.
- **A2: Spatial planning.** Participants can adapt the starter arrangement through their own placement decisions while keeping the central aisle clear.
- **A3: Physical customisation.** Participants can independently remove furniture, change the theme, and adjust lighting within the furnished space.
- **A4: Interaction guidance.** The Grab/Rotate hints and Event Planning Guide help participants understand unfamiliar interactions or recover from difficulties.

The usefulness of the starter furniture and environmental details will be investigated through observed decisions and participant explanations. Their presence alone does not demonstrate improvement.

4. Validation Criteria

Outcome	Score	Definition
Pass	2	Completed independently, including self-directed Guide use
Assist	1	Completed after facilitator instruction or demonstration
Fail	0	Not completed after facilitator assistance

Record technical interruptions or unattempted interactions separately rather than treating them as user failures. Evaluate each assumption using the following evidence:

Assumption	Task and evidence	Support criterion
A1	Task 1, time to Start activation and first relevant planning action	Start activated and a relevant action initiated independently within 30 seconds. Successful furniture manipulation is not required
A2	Task 2, completed edits, placement choices, overlaps, and explanation	All requested edits completed independently, with no visible furniture intersections or blocked aisle, plus a reason for the chosen layout
A3	Tasks 3–4, removal, lighting, and theme outcomes	Each interaction completed independently, scored separately
A4	Tasks 1–4, observed hint/Guide use, subsequent action, and feedback	Documented guidance use followed by the relevant independent action and an explanation of what helped

For A1–A3, use **at least 4/5 independent successes** as validated within this sample, **3/5** as partially validated, and **2/5 or fewer** as requiring further iteration. Report actual counts if the sample differs.

For A4, report how many participants used each support and whether it helped. Opening the Guide alone does not demonstrate its usefulness. Non-use leaves usefulness untested. Completion alone does not establish that guidance or environmental details caused success.

5. Testing Procedure

Method and setup

Use task-based usability testing, observation, brief think-aloud, and post-test questions. Initial unguided exploration reveals starting uncertainty before structured tasks provide

direction. Shared task goals make outcomes comparable, while observations and comments explain difficulties behind the scores.

Test with at least five peers in class, including tutor observation or participation where available. Record previous VR experience and familiarity with the prototype.

Before each participant, restore the Welcome screen, four tables, eight chairs, stage, default lighting and theme, original wand positions, and empty pedestal. Remove furniture added during the previous session. Check the menu, spawning, grabbing, rotation, hints, Remove Zone, lever, wands, and Guide on the Meta Quest. Prepare a timer, results sheet, and mirrored headset view.

Introduction

Approximately 30 seconds

“Thanks for testing XR Event Planning. This is a test of the prototype, not you. Please say what you are looking for or expecting when comfortable. For the first 30 seconds, I’ll let you explore without guidance. Afterwards, I’ll give you some planning tasks. You can pause or stop whenever you need to. Is it okay if I take notes?”

Ask separately for recording consent if recording. Explain safe headset use and basic controller handling without demonstrating application controls.

Assistance rules

During the full initial **30 seconds**, provide no hints, demonstrations, or control explanations, even if help is requested. Record requests and intervene only for safety. Self-discovered Guide use remains independent.

After that period, if Task 1 is incomplete, give one minimal instruction and allow approximately 15 seconds to attempt it. Score Assist if completed or Fail if still incomplete.

For Tasks 2–4, provide help only when requested or after 20 seconds without progress. Give one instruction about the blocked interaction, then demonstrate only if it remains unclear. Record the exact assistance and score only the affected interaction accordingly. Task durations are estimates, with brief flexibility within the six-minute session.

Task 1: Explore and Begin Planning

Approximately 45 seconds

“Imagine you are preparing this venue for a small event. Explore the experience and show me what you would do first.”

Completion: Independently activate Start and initiate a relevant planning action within the initial 30 seconds. This means selecting a furniture option, opening the Guide, or making an observable attempt to grab or rotate existing furniture. Looking around alone does not meet this criterion.

Observe: Start discovery, time to Start activation, first relevant action and its time, hesitation, help requests, and self-directed Guide use. Start timing when the Welcome screen appears after the instruction. An unsuccessful grab can still demonstrate orientation. Furniture manipulation success is assessed in Task 2.

Task 2: Adjust the Venue Layout

Approximately 100 seconds

“Your event needs a small lounge area beside the central aisle. Add a sofa and a plant, move an existing chair, and turn one object to suit your layout. Choose their positions and keep the aisle clear.”

Completion: Sofa and Plant are added, an existing chair is repositioned, and an object is visibly rotated. Furniture does not visibly intersect other objects or obstruct the aisle. Ask briefly, “Why did you choose this arrangement?”

Observe: Placement decisions, menu discovery, spawn positions, failed grabs, rotation attempts, collisions, corrections, hints, Guide use, and awkward physical reach. Record each requested edit separately in the notes if the overall task is incomplete.

Task 3: Remove Furniture

Approximately 35 seconds

“One furniture item is no longer needed for your revised layout. Choose one and remove it.”

Completion: The selected item is removed through the physical Remove Zone.

Observe: Zone discovery, transporting and releasing the object, accidental removal, searching, physical difficulty, and assistance.

Task 4: Customise the Atmosphere

Approximately 50 seconds

“Choose a different theme for your event, then make the venue noticeably brighter or darker.”

Completion: A different theme is applied using a wand and pedestal, and brightness changes using the lever. Score these separately.

Observe: Control discovery, wand placement, lever handling, visible feedback, movement between controls, discomfort, and assistance.

6. Post-Test Questions

Approximately 60 seconds

1. What helped or confused you when deciding where to begin?
2. How did the starting furniture and room details affect your layout choices, if at all?
3. Which interaction was hardest, physically or otherwise, and why?
4. Did you notice or use the hints or Guide? If used, what did they help you do?
5. What one change would most improve planning your event?

Use short follow-ups to clarify observed incidents.

7. Data Collection

Complete one row per participant. Enter 2, 1, or 0 for each outcome. For orientation, record seconds to Start activation and the first relevant planning action, or “not reached within 30 seconds.”

ID	Layout	Remove	Lighting	Theme	Guide Task
P1					
P2					
P3					
P4					
P5					

Additional Observation Notes

For each participant, record:

- VR experience and recording consent.
- Difficulties, unexpected interactions, and how the participant responded.
- Grab/Rotate hint and Guide use, including whether they helped.
- Exact facilitator assistance and which task required it.
- Relevant participant comments, including feedback on the starting layout and VR interactions.
- Technical issues or discomfort.

Keep observations separate from participant comments. After testing, summarise task completion, recurring difficulties, and feedback to identify which design decisions were supported or need improvement. Upload anonymised testing results to GitHub.

Evaluation Against Design Principles

The prototype will also be evaluated against the five design principles from the course lectures. Each principle will be reviewed using relevant prototype features, testing observations, and possible improvements. Any principle that does not apply will be explained.